

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

**COURSE NAME: ARTIFICIAL
INTELLIGENCE AND EXPERT
SYSTEMS**

**COURSE OUTCOME COVERED
COURSE CODE: MLA01**

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.(BTL 3)*

Assessment Tool Weightage: 15%

Total Marks: 100 Time: 60 Mins No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I Kamalli shree V(192525030) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weight	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

VEHICLE FAULT DIAGNOSIS

1. Problem Analysis and Knowledge Identification

Problem Statement:

An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Based on the given rules, the system must determine whether Car101 has a battery problem and whether it should be marked for service.

Given Facts:

- Car101 does not start.
- Car101 has dim headlights.

Given Rules:

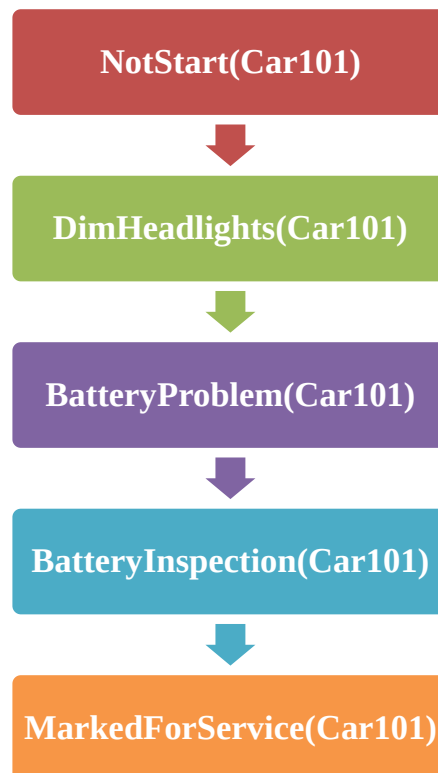
- Vehicles that do not start and have dim headlights may have a battery problem.
- Vehicles with a battery problem require battery inspection.

- Vehicles requiring battery inspection are marked for service.

Entities:

- Vehicle
- Car101
- Battery
- Battery inspection
- Service

Main Inferences:



2. Propositional Logic Representation

Let:

- P = Car101 does not start
- D = Car101 has dim headlights
- B = Car101 has a battery problem
- I = Car101 requires battery inspection
- S = Car101 is marked for service

Facts:

- P
- D

Rules:

- $P \wedge D \rightarrow B$
- $B \rightarrow I$
- $I \rightarrow S$

Therefore:

- $P \wedge D \rightarrow B \rightarrow I \rightarrow S$

Final Propositional Conclusion:

BAIAS

Therefore, Car101 has a battery problem, requires battery inspection, and should be marked for service.

For this logic exercise, the statement “may have a battery problem” is represented as an implication so that the requested inference can be performed.

3. Predicates, Variables, Constants and Quantifiers

Predicates:

NotStart(x) - Vehicle x does not start.

DimHeadlights(x) - Vehicle x has dim headlights.

BatteryProblem(x) - Vehicle x has a battery problem.

BatteryInspection(x) - Vehicle x requires battery inspection.

MarkedForService(x) - Vehicle x is marked for service.

Constant : Car101

Variable : x , represents any vehicle.

Quantifier:

- The general rules use the universal quantifier: $\forall x$ meaning "for every vehicle x ."

4. First-Order Logic Representation

Facts:

- $\text{NotStart}(\text{Car101})$
- $\text{DimHeadlights}(\text{Car101})$

Rule 1:

- $\forall x[\text{NotStart}(x) \wedge \text{DimHeadlights}(x) \rightarrow \text{BatteryProblem}(x)]$

Rule 2:

- $\forall x[\text{BatteryProblem}(x) \rightarrow \text{BatteryInspection}(x)]$

Rule 3:

- $\forall x[\text{BatteryInspection}(x) \rightarrow \text{MarkedForService}(x)]$

5. Logical Connectives

The main connectives used are:

a) AND ($\wedge$):

Both conditions must be true:

- $\text{NotStart}(x) \wedge \text{DimHeadlights}(x)$

b) Implication ($\rightarrow$):

- One condition leads to another:

$$\text{BatteryProblem}(x) \rightarrow \text{BatteryInspection}(x)$$

Complete Formula:

- $[\text{NotStart}(x) \wedge \text{DimHeadlights}(x)] \rightarrow \text{BatteryProblem}(x)$
- $\text{BatteryProblem}(x) \rightarrow \text{BatteryInspection}(x)$
- $\text{BatteryInspection}(x) \rightarrow \text{MarkedForService}(x)$

6. Unification and Substitution

Consider Rule 1:

- $\text{NotStart}(x) \wedge \text{DimHeadlights}(x) \rightarrow \text{BatteryProblem}(x)$

Known facts:

- $\text{NotStart}(\text{Car101})$
- $\text{DimHeadlights}(\text{Car101})$

Unification:

Match: $\text{NotStart}(x)$ with $\text{NotStart}(\text{Car101})$

The substitution is: $\theta = \{x/\text{Car101}\}$

The second predicate also matches:

- $\text{DimHeadlights}(x)$ With $\text{DimHeadlights}(\text{Car101})$

Therefore: $x = \text{Car101}$

After Substitution:

- $\text{NotStart}(\text{Car101}) \wedge \text{DimHeadlights}(\text{Car101})$

therefore:

$\text{BatteryProblem}(\text{Car101})$
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7. Step-by-Step Inference

Step 1: Initial Facts

- $\text{NotStart}(\text{Car101})$
- $\text{DimHeadlights}(\text{Car101})$

Step 2: Apply Rule 1

Both conditions are true:

- $\text{NotStart}(\text{Car101}) \wedge \text{DimHeadlights}(\text{Car101})$

Therefore:

- $\text{BatteryProblem}(\text{Car101})$

Step 3 : Apply Rule 2

Therefore:

- $\text{BatteryInspection}(\text{Car101})$

Step 4 : Apply Rule 3

Therefore:

- $\text{MarkedForService}(\text{Car101})$

8. Resolution Proof

To prove that Car101 has a battery problem, start with the relevant rule:

- $\text{NotStart}(x) \wedge \text{DimHeadlights}(x) \rightarrow \text{BatteryProblem}(x)$

Convert it into clause form:

- $\neg \text{NotStart}(x) \vee \neg \text{DimHeadlights}(x) \vee \text{BatteryProblem}(x)$

Substitute:

- $x = \text{Car101}$

giving:

- $\neg \text{NotStart}(\text{Car101}) \vee \neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$

Given facts:

- $\text{NotStart}(\text{Car101})$
- $\text{DimHeadlights}(\text{Car101})$

Resolution 1:

Resolve:

- $\neg \text{NotStart}(\text{Car101}) \vee \neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$ With $\text{NotStart}(\text{Car101})$

Result:

- $\neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$

Resolution 2:

Resolve with:

- $\text{DimHeadlights}(\text{Car101})$

Result:

- $\text{BatteryProblem}(\text{Car101})$

Therefore:

- Car101 has a battery problem

9. Resolution for Service

Rule 2:

- $\text{BatteryProblem}(x) \rightarrow \text{BatteryInspection}(x)$

Clause:

- $\neg \text{BatteryProblem}(x) \vee \text{BatteryInspection}(x)$

Using:

- $\text{BatteryProblem}(\text{Car101})$

we obtain:

- $\text{BatteryInspection}(\text{Car101})$

Rule 3:

- $\text{BatteryInspection}(x) \rightarrow \text{MarkedForService}(x)$

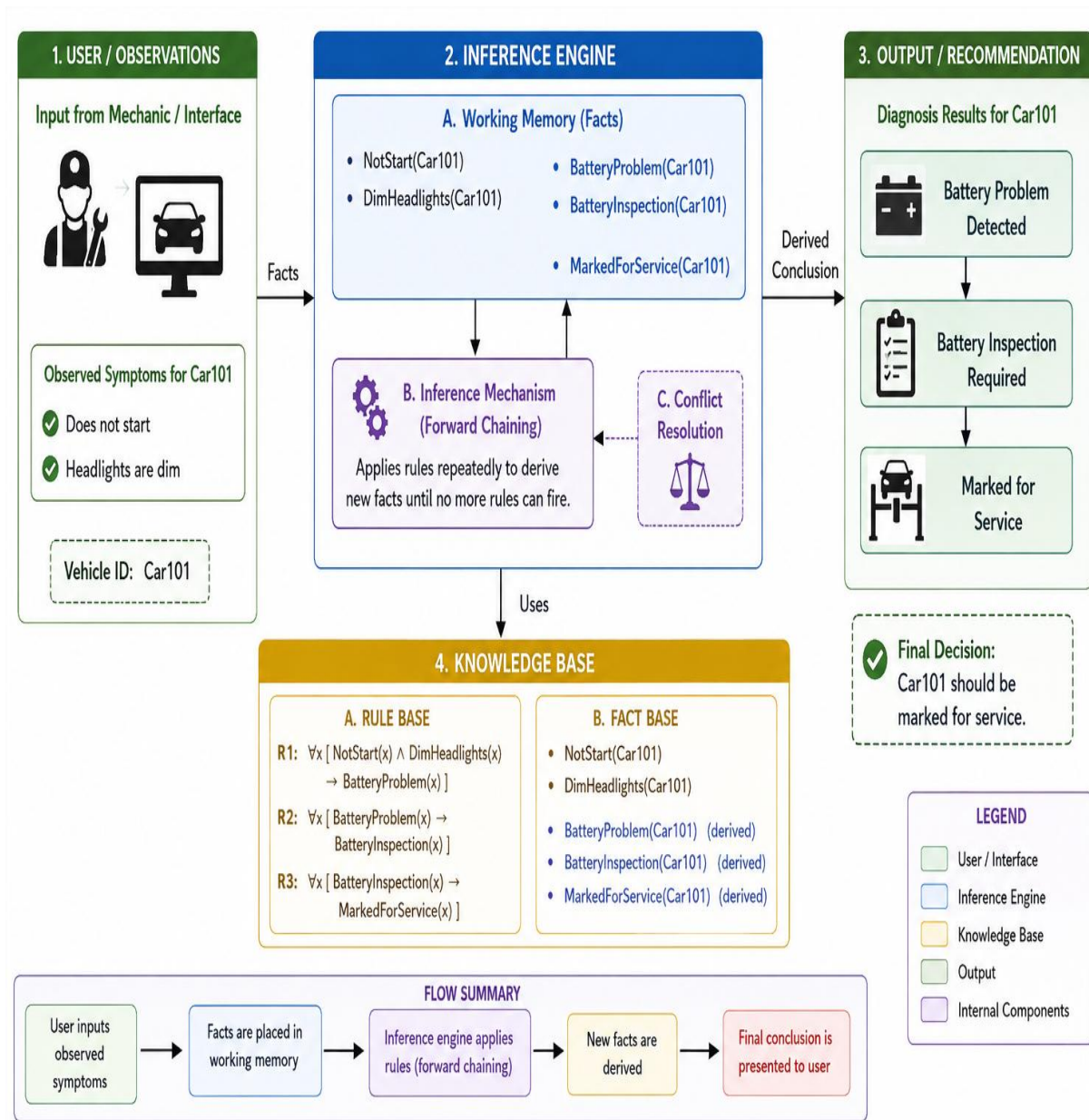
Clause:

- $\neg \text{BatteryInspection}(x) \vee \text{MarkedForService}(x)$

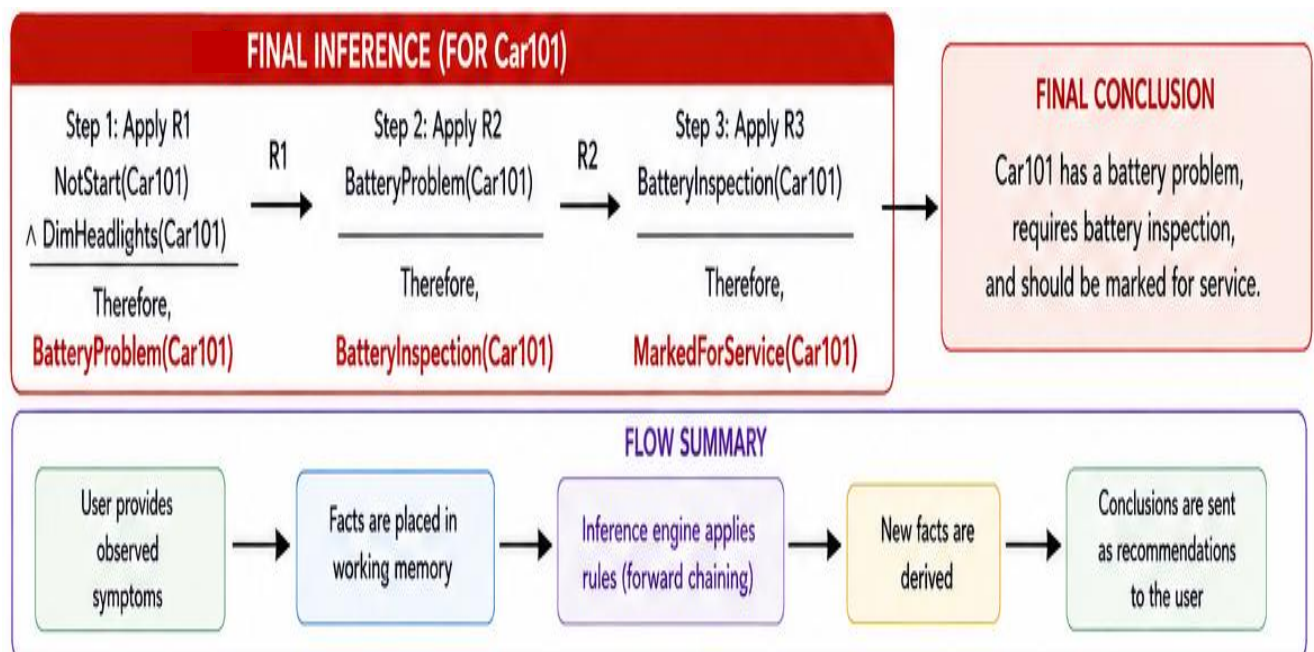
we obtain:

- $\text{MarkedForService}(\text{Car101})$

10. System Architecture



11. Final Inference



12. Advantages

- Provides a clear and systematic diagnosis process.
- Uses explicit rules that are easy to understand and verify.
- FOL allows the rules to be applied to different vehicles.
- Unification allows vehicle-specific facts to match general rules.
- Resolution provides a formal method for verifying the inference.

13. Limitations

- The diagnosis depends on the accuracy of the given facts.
- The system considers only faults represented in its knowledge base.
- Multiple vehicle faults may produce similar symptoms.
- The rule-based system does not represent uncertainty directly.
- New vehicle faults require additional rules to be added.

14. Conclusion

The expert system successfully uses logical reasoning to diagnose the fault in Car101. The two observed symptoms—the vehicle does not start and its headlights are dim—satisfy the conditions of the first rule, allowing the system to infer a battery problem. This conclusion triggers the second rule, which determines that battery inspection is required. The third rule then marks Car101 for service.

The use of Propositional Logic and First-Order Logic provides a structured representation of the knowledge. Unification and substitution allow the general rules to be applied specifically to Car101, while resolution formally derives the battery-problem conclusion from the given facts. The complete reasoning chain demonstrates how an expert system can move from observed symptoms to a service decision in a transparent and explainable manner.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Analysis and Understanding	15%	13–15% – Clearly understands and analyzes the real-world problem, objectives, entities, facts, constraints, and expected inference.	10–12% – Understands the problem with minor omissions.	7–9% – Shows partial understanding with some important elements missing.	0–6% – Poor or incorrect understanding of the problem.
Knowledge Engineering & Knowledge Base Design	15%	13–15% – Develops a complete, consistent, and well-structured knowledge base with appropriate facts, predicates, entities, and rules.	10–12% – Knowledge base is well designed with minor inconsistencies or omissions.	7–9% – Basic knowledge base is developed but several facts/rules are missing or unclear.	0–6% – Knowledge base is incomplete, inconsistent, or inappropriate.
Propositional & First-Order Logic Representation	20%	17–20% – Correctly represents the domain using propositions, predicates, variables, constants, quantifiers, and logical connectives with excellent accuracy.	13–16% – Logic representation is mostly correct with minor errors.	9–12% – Provides basic representations but has several logical or syntactic errors.	0–8% – Logic representation is incorrect, incomplete, or missing.
Unification & Resolution	15%	13–15% – Correctly performs substitutions, identifies the MGU, and applies resolution systematically to derive valid conclusions.	10–12% – Applies unification and resolution correctly with minor errors.	7–9% – Demonstrates partial understanding but misses important steps or contains errors.	0–6% – Unable to correctly apply unification or resolution.

Forward Chaining & Backward Chaining	15%	13–15% – Correctly applies both forward and backward chaining with clear step-by-step reasoning and appropriate rule selection.	10–12% – Correctly applies both methods with minor errors or omissions.	7–9% – Demonstrates partial understanding of one or both chaining methods.	0–6% – Incorrectly applies or fails to demonstrate chaining techniques.
Inference & Reasoning Accuracy	10%	9–10% – Produces logically valid conclusions and clearly explains how facts and rules lead to the final inference.	7–8% – Conclusions are mostly correct with minor reasoning gaps.	5–6% – Some correct inferences but reasoning is incomplete or unclear.	0–4% – Conclusions are incorrect, unsupported, or missing.
Knowledge Representation Diagram / System Architecture	5%	5% – Provides a complete, accurate, well-labelled diagram showing knowledge base, inference engine, reasoning process, and output.	4% – Diagram is mostly correct with minor omissions.	3% – Diagram is partially complete or lacks clarity.	0–2% – Diagram is missing or incorrect.
Originality, Critical Thinking & Presentation	5%	5% – Demonstrates excellent independent thinking, appropriate real-world application, comparison of reasoning approaches, and clear presentation.	4% – Demonstrates good reasoning and clear presentation.	3% – Shows limited critical thinking and basic presentation.	0–2% – Shows little originality, weak reasoning, or poor presentation.